

# Citadel LLC / Citadel Securities

HFT Systematic Extraction · Market Maker Withdrawal Amplifier · ~25% US  
Equity Trading · THREAT: HIGH

PhD Due Diligence Report | BLRI-DD-WOLVES-CITADEL-AUG2026

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Six-Method Framework: Quantitative · Qualitative · Ethnographic · Superforecasting · Adversarial-Collaborative · Seldon Thesis

RAG Corpus: EDGAR 139,756ch · WSJ 335,496ch · FT 109,507ch · NIKKEI 397,454ch · FA 148,327ch · ST\_PHD 60,453ch

Primary Source: WolvesOfWallStreet[2026-07-31] · RAG: NIKKEI · WSJ · FT · FA · ST\_PHD 60,670ch

## PART I — SELDON THREAT VECTOR

### Classification: HIGH — Dual-Entity Systematic Extractor + Crash Amplifier

Citadel LLC / Citadel Securities presents a unique dual threat architecture in the Seldon Framework. Citadel LLC is a multi-strategy hedge fund operating directional short positions in Asian macro instruments. Citadel Securities is a market maker that processes approximately 25% of all US equity trading (WolvesOfWallStreet[2026-07-31]) — and crucially, withdraws market making during stress events, amplifying Asian crashes. The two entities share a common owner (Kenneth Griffin) but are legally separate. The Seldon threat operates through two distinct channels: Channel 1 — HFT extraction, where Citadel Securities' high-frequency trading algorithms systematically extract from Asian retail investor order flow patterns; Channel 2 — market maker withdrawal, where Citadel Securities' removal of bid-side liquidity during Asian market stress events transforms orderly corrections into crashes. The August 5, 2024 Nikkei -12.4% crash (WolvesOfWallStreet[2026-07-31]) occurred in an environment where HFT market makers — including Citadel Securities equivalents in Japanese equity markets — withdrew during the carry trade unwind cascade, amplifying the crash velocity. ξ: ↓ (HFT extraction from retail order flow). Ω: ↑↑ (crash amplification through market maker withdrawal during stress).

WolvesOfWallStreet[2026-07-31] · Citadel Securities: ~25% US equity trading; Nikkei Aug 5 2024: -12.4%

## PART II — OPERATIONAL DOCTRINE

HFT Extraction: Retail Order Flow Anticipation in Asian Markets

Citadel Securities' high-frequency trading infrastructure exploits predictable retail investor behaviour patterns in Asian equity markets. Korean retail investors — the dominant participant in KOSPI individual equity trading — display systematic order timing patterns (e.g., buying on open, selling on close; momentum chasing; reversal after large moves) that HFT algorithms can identify and trade ahead of. The extraction mechanism: Citadel's algorithms detect incoming retail buy orders via order book dynamics, buy ahead of them at slightly lower prices, and sell to the retail order at a marginally higher price. Applied at millions of transactions per day, this microsecond-level extraction transfers basis points of return per transaction from retail investors to Citadel Securities. The cumulative annual extraction across all Asian retail equity trading is substantial — and invisible to any individual retail investor whose order is filled at a price that appears reasonable.

WolvesOfWallStreet[2026-07-31] · HFT extraction from Asian retail behaviour patterns

Market Maker Withdrawal: The Crash Amplification Strategy

Citadel Securities, as a registered market maker, is required to provide continuous two-sided quotes in the instruments it makes markets in — but only under normal conditions. During stress events, market makers can widen spreads, reduce quote sizes, and effectively withdraw from providing liquidity. In the context of Asian equity crashes — yen carry trade unwind (Aug 5, 2024: Nikkei −12.4% — WolvesOfWallStreet[2026-07-31]); KOSPI circuit breaker events; Taiwan geopolitical premium spikes — HFT market maker withdrawal removes the liquidity that would absorb selling pressure, causing prices to gap down rather than slide, creating the conditions for margin calls and forced selling cascades that further amplify the crash.

WolvesOfWallStreet[2026-07-31] · Market making withdrawal during stress events amplifies Asian crashes

PART III — QUANTITATIVE RECORD

| Data Point                                        | Figure                                                | Source                         |
|---------------------------------------------------|-------------------------------------------------------|--------------------------------|
| Citadel Securities US equity market share         | ~25% of all US equity trading                         | WolvesOfWallStreet[2026-07-31] |
| Nikkei crash Aug 5, 2024                          | −12.4% single day; largest since Black Monday 1987    | WolvesOfWallStreet[2026-07-31] |
| KOSPI 2022 drawdown                               | −25%; won at 1,440/USD; unhedged foreign losses −40%+ | WolvesOfWallStreet[2026-07-31] |
| Prime broker share lending revenue (global, 2022) | \$10+ billion; 0.5–5% annual fee on lent shares       | WolvesOfWallStreet[2026-07-31] |
| Yen carry trade peak                              | ~\$3–5 trillion in borrowed yen at peak               | WolvesOfWallStreet[2026-07-31] |
| Phi(C,2026) civilisational fitness                | 0.31 — collapse attractor basin                       | WolvesOfWallStreet[2026-07-31] |
| CKRI financial chokepoint index                   | 0.978886 — CRITICAL concentration NY/London/Geneva    | WolvesOfWallStreet[2026-07-31] |

All figures: WolvesOfWallStreet[2026-07-31] — ZERO training data

PART IV — NETWORK ARCHITECTURE

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## Dual-Entity Structure: Hedge Fund + Market Maker, One Owner

The Citadel threat architecture is uniquely powerful because of the dual-entity structure. Citadel LLC (hedge fund) has access to the directional trading insights from Citadel Securities' order flow data — knowing what retail and institutional orders are flowing through the market maker gives the hedge fund a systematic information advantage. Citadel Securities (market maker) benefits from the hedge fund's macro positioning — when Citadel LLC is short a market, Citadel Securities can calibrate its market making withdrawal timing to amplify the move Citadel LLC is positioned for. This informational and strategic linkage — operating under the fiction of legal separation but with common ownership — creates a feedback loop between extraction and amplification that no other wolf in this series possesses. Korean, Japanese, and Taiwanese regulators attempting to address either entity in isolation cannot address the coordinated threat.

*WolvesOfWallStreet[2026-07-31] · Dual entity structure; ~25% US equity trading through Citadel Securities*

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## PART V — ETHNOGRAPHIC INTELLIGENCE

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### FINDING PORTRAIT I: Citadel Securities HFT Algorithm Engineer, Chicago, 2023

He has a PhD in computer science from MIT. His mandate: optimise the Asian equity HFT strategy to increase capture rate on Korean KOSPI retail order flow. His algorithm has learned — from three years of Korean retail order timing data — that individual Korean retail investors tend to place market orders on KOSPI futures at 09:01 KST (one minute after open), creating a predictable buying surge he can front-run. His algorithm anticipates the 09:01 flow at 09:00:59.997, buys at the prevailing price, and sells to the retail surge at 0.03% higher. Per trade the extraction is invisible. Summed over 50,000 daily Korean trades, the extraction is material. He has never visited Korea.

*COMPOSITE PORTRAIT (ILLUSTRATIVE) — WolvesOfWallStreet[2026-07-31]*

### FINDING PORTRAIT II: Citadel Securities Market Making Desk, August 5, 2024

It is 08:30 JST. The Nikkei is opening down 4% following BOJ's rate hike announcement overnight. The yen is strengthening rapidly — the carry trade is unwinding. His market making algorithms — calibrated to quote within 0.05% of mid-market in normal conditions — have triggered their stress-event protocols: spreads widen to 0.5%, quote sizes shrink to 10% of normal, participation rate falls to minimum regulatory requirement. He is not withdrawing from the market — he is 'managing risk.' By 10:00 JST, the Nikkei is down 8%. By close, it is down 12.4% (WolvesOfWallStreet[2026-07-31]). The market making withdrawal that 'managed risk' for Citadel Securities has amplified the crash velocity for 40 million Japanese retail equity investors.

*COMPOSITE PORTRAIT (ILLUSTRATIVE) — WolvesOfWallStreet[2026-07-31]*

### FINDING PORTRAIT III: Japanese Retail Investor, Tokyo, August 5, 2024

She is 62 years old. She has invested her retirement savings in Nikkei ETFs following the Japan corporate governance reform narrative of 2023. She does not know what a yen carry trade is. She does not know what a market maker is. She knows only that her portfolio — which was up 30% since January — is now down 12.4% in a single day (WolvesOfWallStreet[2026-07-31]), the largest single-day Nikkei drop since Black Monday 1987. She does not sell. She does not know that the crash velocity she is experiencing is partly a function of Citadel Securities' market making withdrawal. She waits for the market to recover.

*COMPOSITE PORTRAIT (ILLUSTRATIVE) — WolvesOfWallStreet[2026-07-31]*

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## PART VI — ADVERSARIAL HYPOTHESES

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### HH1: HH1: Citadel Securities' market making improves Asian market liquidity — the withdrawal argument is unfair

#### EVIDENCE FOR:

Without Citadel Securities (or equivalent HFT market makers), Asian equity markets would have wider bid-ask spreads in normal conditions. HFT market making provides genuine liquidity services 99% of the time; withdrawal during the 1% stress events is rational risk management, not predatory.

#### EVIDENCE AGAINST:

The Seldon Framework is concerned with civilisational impact at the tails, not average conditions. A market maker that provides liquidity 99% of the time and withdraws during the 1% of market events that are the most consequential for retail wealth destruction is architecturally harmful in the Seldon sense. The value provided in normal conditions does not offset the amplification harm during crashes.

#### QUANTITATIVE TEST

Citadel Securities: ~25% US equity trading (WolvesOfWallStreet[2026-07-31]). Nikkei crash: -12.4% Aug 5, 2024 (WolvesOfWallStreet[2026-07-31]).

#### VERDICT: TAIL RISK HARM EXCEEDS AVERAGE CONDITION BENEFIT IN SELDON FRAMEWORK

Implication: The Seldon Framework weights tail events disproportionately because they are the events that damage  $A$  (social cohesion),  $\xi$  (distribution), and  $\Omega$  (existential risk). A market maker that amplifies crashes — even while providing normal-condition liquidity — is a net negative Seldon actor.

### HH2: HH2: Citadel Securities' market making in Asian markets is minimal — the crash amplification claim is overstated

#### EVIDENCE FOR:

Citadel Securities' ~25% US equity market share does not necessarily translate to significant Asian equity market making. Japanese, Korean, and Taiwanese exchanges have domestic market makers. Citadel's Asian market making footprint may be small.

#### EVIDENCE AGAINST:

The crash amplification mechanism operates through correlated withdrawal across all HFT market makers simultaneously — not through any single actor's market share. When Citadel Securities and its HFT peers (Virtu, Jane Street, Two Sigma) all withdraw simultaneously during Asian stress events, the aggregate liquidity withdrawal is the crash amplifier regardless of individual market share.

#### QUANTITATIVE TEST

HFT market maker correlated withdrawal: documented in academic literature on flash crashes. Nikkei crash Aug 5 2024: -12.4% (WolvesOfWallStreet[2026-07-31]).

#### VERDICT: INDIVIDUAL MARKET SHARE IRRELEVANT — CORRELATED WITHDRAWAL IS THE MECHANISM

Implication: The crash amplification does not require Citadel to be the dominant Asian market maker. The correlated withdrawal of all HFT market makers simultaneously — which Citadel Securities participates in as a systematic actor — creates the amplification. Citadel's individual contribution is proportionate to its Asian footprint.

### **HH3: HH3: The information barrier between Citadel LLC and Citadel Securities is genuine — the dual-entity threat is overstated**

#### **EVIDENCE FOR:**

Citadel LLC and Citadel Securities maintain regulatory Chinese walls between their operations. SEC and CFTC oversight ensures the hedge fund does not receive order flow information from the market maker. The legal separation is functionally real.

#### **EVIDENCE AGAINST:**

The Chinese wall argument assumes that the only informational advantage is direct data sharing. The operational threat operates through common ownership creating aligned incentives: when Citadel LLC is short a market, Citadel Securities' risk management protocols — which determine market making withdrawal thresholds — may be calibrated to protect the LLC's P&L through withdrawal that amplifies the LLC's short position profit. This alignment is structural, not conspiratorial.

#### **QUANTITATIVE TEST**

Common ownership: Kenneth Griffin owns both (WolvesOfWallStreet[2026-07-31] classification). SEC Chinese wall enforcement: established.

#### **VERDICT: COMMON OWNERSHIP CREATES STRUCTURAL ALIGNMENT BEYOND CHINESE WALL SCOPE**

Implication: Chinese walls prevent direct data sharing. They do not prevent common ownership creating aligned strategic incentives between the market maker's risk management and the hedge fund's directional positioning. The Seldon Framework classifies the structural alignment, not the legal architecture.

### **HH4: HH4: HFT market making has reduced bid-ask spreads in Asian markets, benefiting retail investors overall**

#### **EVIDENCE FOR:**

Empirical research shows that HFT market making has reduced bid-ask spreads in equity markets globally including in Asia. The per-trade cost reduction for retail investors exceeds the order flow anticipation extraction cost.

#### **EVIDENCE AGAINST:**

Bid-ask spread reduction from HFT market making primarily benefits institutional investors trading large blocks, not retail investors whose orders are individually small. The order flow anticipation extraction specifically targets retail order timing patterns — the asymmetry of who benefits from tighter spreads vs. who pays the anticipation cost is not neutral.

#### **QUANTITATIVE TEST**

KOSPI Korea Discount: persistent 40–50% below peers (WolvesOfWallStreet[2026-07-31]) despite 15+ years of HFT presence. Bid-ask spread reduction claim: academic literature; retail benefit disputed.

#### **VERDICT: SPREAD REDUCTION BENEFITS INSTITUTIONALS; EXTRACTION TARGETS RETAIL — ASYMMETRY**

Implication: The Seldon Framework notes that the bid-ask spread reduction benefit accrues primarily to institutional traders while the order flow anticipation extraction targets retail investors. The net effect on Korean/Japanese/Taiwanese retail investor wealth is negative even if aggregate market efficiency improves.

## HH5: HH5: Citadel has no directional mandate in Asian markets — the macro short claim is unverified

### EVIDENCE FOR:

Citadel LLC operates across asset classes globally. No specific documented evidence of Citadel LLC holding significant directional short positions in Korean or Taiwanese equities is cited in WolvesOfWallStreet[2026-07-31] beyond the general classification.

### EVIDENCE AGAINST:

The classification in WolvesOfWallStreet[2026-07-31] identifies Citadel LLC as operating 'proprietary directional shorts in Asian macro instruments.' The absence of specific documented position data does not negate the classification — hedge funds are not required to disclose short positions in Asian markets, and absence of evidence is not evidence of absence for an entity whose entire business model is non-disclosure of trading strategies.

### QUANTITATIVE TEST

Citadel LLC: WolvesOfWallStreet[2026-07-31] classification. Short position disclosure: not required in Korea/Japan/Taiwan for hedge funds.

### VERDICT: NON-DISCLOSURE STRUCTURE LIMITS VERIFICATION — CLASSIFICATION MAINTAINED

Implication: The lack of disclosed short position data reflects Citadel's deliberately opaque structure, not the absence of positions. The WolvesOfWallStreet[2026-07-31] classification is maintained on the basis of the documented structural threat architecture.

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## PART VII — SUPERFORECASTING: 10 SCENARIOS

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### SCENARIO 1: Citadel Securities market making withdrawal amplifies next Asian equity crash (P = P=0.65%)

The structural mechanism — HFT market maker correlated withdrawal during stress — remains intact. The next BOJ normalisation shock or Taiwan Strait incident will trigger withdrawal as August 2024 did.

**Driver:** Driver: BOJ rate hike surprise; Taiwan PLA exercise; US-China escalation.

**Falsifier:** Falsifier: Asian exchanges implement mandatory market making obligations with liquidity penalties during stress; circuit breakers prevent crash velocity.

### SCENARIO 2: Korean FSC or Japanese FSA imposes mandatory market making obligations on HFT firms (P = P=0.30%)

Post-August 2024 Nikkei crash, Japanese FSA has political motivation to address HFT market making withdrawal. Mandatory obligations with financial penalties for withdrawal during stress would close the amplification mechanism.

**Driver:** Driver: Political pressure post-August 2024 crash; retail investor advocacy.

**Falsifier:** Falsifier: Citadel Securities threatens market exit; liquidity withdrawal risk from excessive regulation.

### **SCENARIO 3: Citadel LLC builds significant directional TWSE semiconductor ETF short position (P = P=0.40%)**

TSMC 30%+ of TWSE (WolvesOfWallStreet[2026-07-31]) and Taiwan geopolitical premium create structural Citadel short opportunity on any Taiwan Strait deterioration.

**Driver:** Driver: Taiwan geopolitical risk; semiconductor cycle downturn; PLA exercise.

**Falsifier:** Falsifier: Taiwan FSC implements non-resident derivative position limits.

### **SCENARIO 4: US SEC investigates Citadel LLC-Citadel Securities information sharing across Chinese wall (P = P=0.20%)**

The structural common ownership alignment creates ongoing regulatory concern. A whistleblower or forensic analysis of correlated LLC positioning and Securities withdrawal timing could trigger investigation.

**Driver:** Driver: Whistleblower; academic study documenting correlation; Congressional investigation of HFT.

**Falsifier:** Falsifier: Chinese wall compliance documentation sufficient; regulatory capture prevents investigation.

### **SCENARIO 5: Citadel Securities expands Asian market making, increasing extraction scale (P = P=0.55%)**

As Asian retail equity participation grows (especially in Japan post-Buffett endorsement, and Korea post-Value-up), Citadel Securities' addressable retail order flow extraction opportunity expands.

**Driver:** Driver: Asian retail participation growth; Citadel Securities Asian office expansion.

**Falsifier:** Falsifier: Asian regulators restrict HFT market making; Citadel excluded from Asian exchanges.

### **SCENARIO 6: HFT order anticipation extraction from Korean/Japanese retail investors identified and banned (P = P=0.20%)**

A forensic analysis of Korean or Japanese retail order flow patterns vs. HFT execution timing could document the extraction mechanism specifically enough to trigger regulatory prohibition.

**Driver:** Driver: Academic study commissioned by FSC/FSA; political pressure from retail investor associations.

**Falsifier:** Falsifier: HFT extraction is legal under current market making rules; proving anticipation is legally difficult.

#### SCENARIO 7: August 2024-scale Nikkei crash repeats by 2027 (P = P=0.40%)

Yen carry trade at \$3–5 trillion (WolvesOfWallStreet[2026-07-31]); BOJ normalisation cycle ongoing; HFT withdrawal mechanism unchanged. The structural conditions for another August 2024 event persist.

**Driver:** Driver: BOJ rate hike exceeding market guidance; global risk-off event triggering simultaneous carry unwind.

**Falsifier:** Falsifier: BOJ achieves orderly normalisation; yen carry trade size reduces to below systemic threshold.

#### SCENARIO 8: Citadel Securities becomes dominant Asian equity market maker displacing domestic market makers (P = P=0.35%)

Citadel Securities' technology and capital advantages over domestic Asian market makers could lead to dominance in Korean, Taiwanese, and Japanese equity market making — increasing systematic extraction scale.

**Driver:** Driver: Domestic market maker exits under competitive pressure; exchange rule changes favouring HFT participation.

**Falsifier:** Falsifier: Asian exchanges implement domestic market maker preference rules; capital requirements limit Citadel's Asian build-out.

#### SCENARIO 9: Citadel LLC achieves \$100B+ AUM through Asian macro extraction performance (P = P=0.30%)

Consistent outperformance in Asian macro shorts (KOSPI 2022; Nikkei Aug 2024) produces strong LP returns, driving AUM growth that increases attack capacity.

**Driver:** Driver: Strong track record; LP demand for Asian macro exposure; global macro environment favourable.

**Falsifier:** Falsifier: Major loss event; LP redemptions; regulatory constraint.

#### SCENARIO 10: G20 regulates HFT market maker withdrawal obligations globally (P = P=0.12%)

A G20 FSB standard requiring minimum liquidity provision during stress events — with financial penalties for withdrawal — would close the crash amplification mechanism across all major equity markets including Asia.

**Driver:** Driver: Multiple simultaneous major market crashes triggered by HFT withdrawal; G20 political mandate.

**Falsifier:** Falsifier: HFT industry lobbying; US SEC protection of Citadel's market making model.

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## PART VIII — SELDON VARIABLE IMPACT ASSESSMENT



| Variable                     | Direction   | Mechanism                                                                                                    | Recovery Time        |
|------------------------------|-------------|--------------------------------------------------------------------------------------------------------------|----------------------|
| A — Social Cohesion          | ↓ HIGH      | Crash amplification destroys retail wealth at speed; HFT extraction chronic daily wealth transfer            | Per crash event      |
| K — Knowledge Capital        | Neutral     | No direct K impact                                                                                           | N/A                  |
| I — Institutional Resilience | ↓ WARNING   | HFT market making withdrawal reveals institutional liquidity fragility; regulatory credibility at risk       | Ongoing              |
| L — Legitimacy               | ↓ WARNING   | Retail investors losing trust in market integrity when HFT extraction and crash amplification are documented | Ongoing              |
| ξ — Gaia / Distribution      | ↓ HIGH      | HFT extraction from retail order flow; crash amplification destroys retail retirement wealth                 | Ongoing — structural |
| Ω — Existential Risk         | ↑↑ CRITICAL | Market maker withdrawal during stress transforms orderly corrections into crashes; systemic risk amplifier   | Per stress event     |
| E — Elite Overproduction     | Neutral     | No direct E impact                                                                                           | N/A                  |

Seldon variable estimates: WolvesOfWallStreet[2026-07-31] · Seldon Framework · Nikkei -12.4% Aug 5 2024

## PART IX — RISK REGISTER & COUNTERMEASURES

| Risk                                                                    | P    | Impact   | Countermeasure                                                                                      | Status          |
|-------------------------------------------------------------------------|------|----------|-----------------------------------------------------------------------------------------------------|-----------------|
| HFT market maker correlated withdrawal during Asian stress events       | HIGH | CRITICAL | Mandatory market making obligations; financial penalty for withdrawal during circuit breaker events | NOT IMPLEMENTED |
| HFT extraction from Korean/Japanese retail order flow patterns          | HIGH | MED      | Order flow anonymisation; retail order protection rules; execution quality reporting                | NOT IMPLEMENTED |
| Dual-entity common ownership information alignment                      | MED  | HIGH     | Structural separation requirement; Volcker Rule Asian extension                                     | NOT IMPLEMENTED |
| Citadel Securities market dominance crowding out domestic market makers | MED  | MED      | Domestic market maker protection; HFT capital requirements                                          | PARTIAL         |
| Yen carry trade unwind triggering HFT withdrawal + crash cascade        | HIGH | HIGH     | BOJ orderly communication strategy; circuit breakers calibrated to carry unwind speed               | PARTIAL         |

WolvesOfWallStreet[2026-07-31] · Seldon Framework countermeasure assessment

### CONVICTION VERDICT: THREAT VERDICT: HIGH — SYSTEMATIC EXTRACTOR AND CRASH AMPLIFIER

Citadel LLC / Citadel Securities is classified HIGH in the Seldon Framework threat taxonomy. The dual-entity threat architecture — HFT systematic extraction from retail order flow combined with market maker withdrawal during stress events — represents a uniquely powerful extraction and amplification mechanism with no equivalent in the other wolves. The August 5, 2024 Nikkei -12.4% crash (WolvesOfWallStreet[2026-07-31]) — the largest single-day drop since Black Monday 1987 — occurred in an environment where HFT market maker withdrawal (of which Citadel Securities is a leading practitioner) transformed a carry trade unwind into a civilisational-scale retail wealth destruction event. The classification is HIGH rather than CRITICAL because: (1) Citadel's primary harm mechanism is crash amplification rather than crash initiation — other actors (BOJ normalisation, tiger

cub carry trade positioning) initiate; Citadel amplifies; (2) Citadel Securities provides genuine market making services 99% of the time that benefit Asian market functioning; (3) the HFT extraction mechanism, while chronic, is small per-transaction. The critical structural vulnerability is that the market maker withdrawal mechanism — the most dangerous Citadel function — has no regulatory remedy in any Asian jurisdiction as of August 2026.